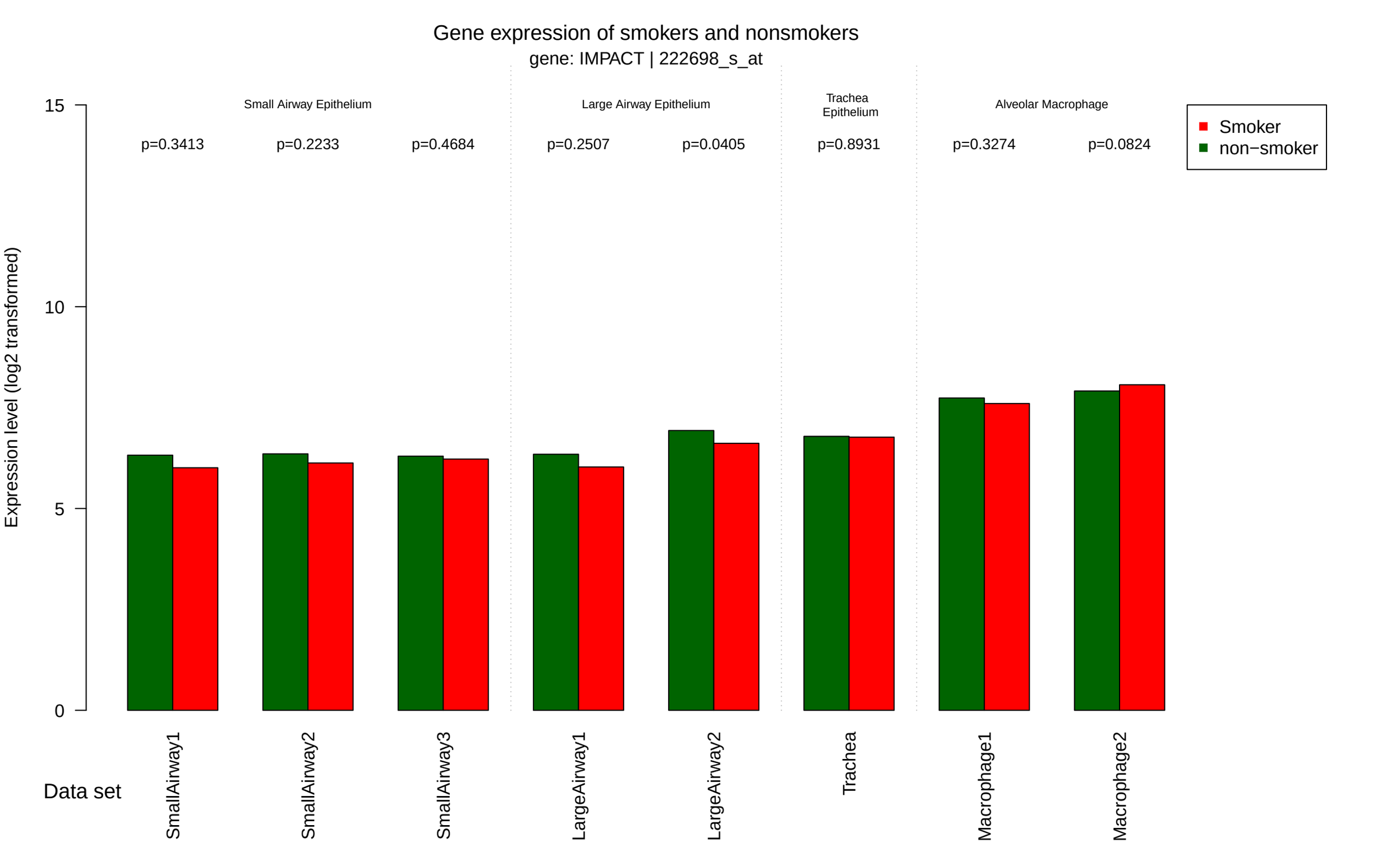
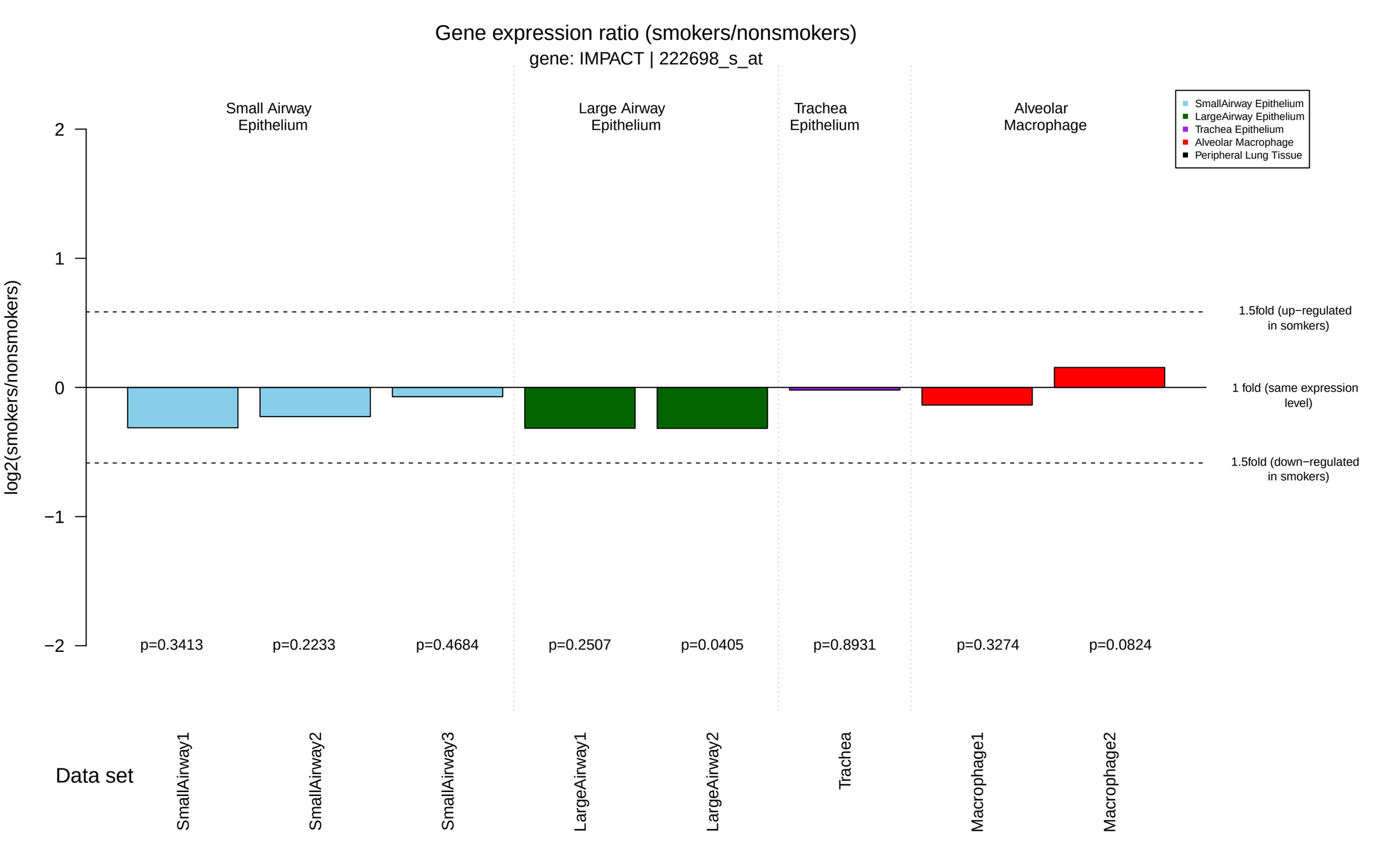
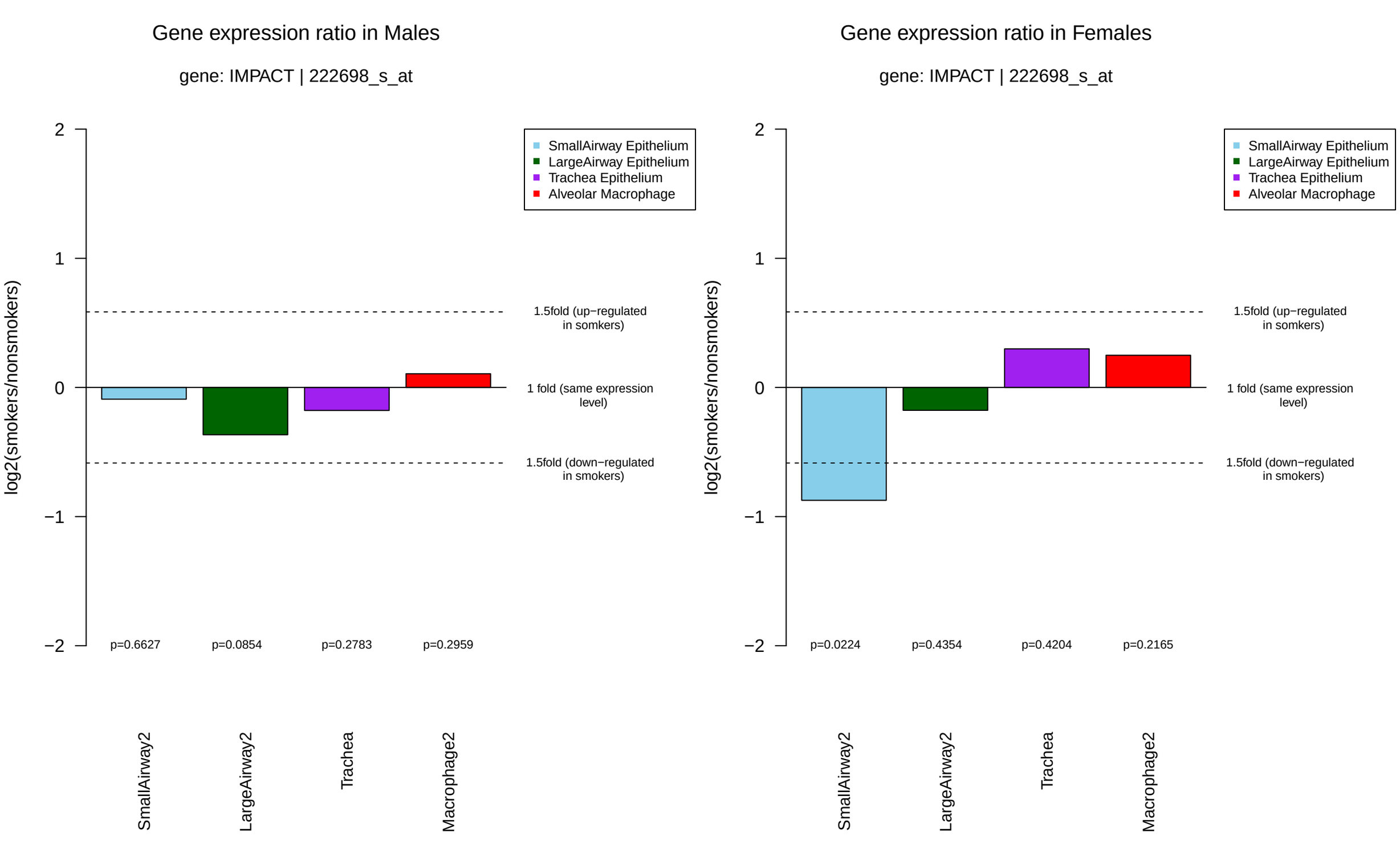
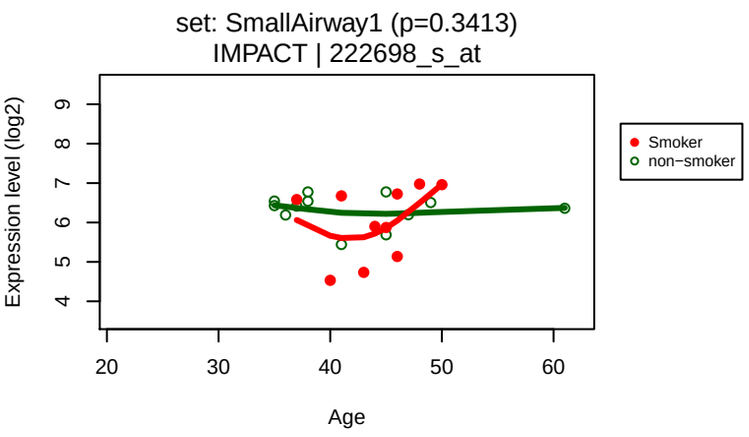
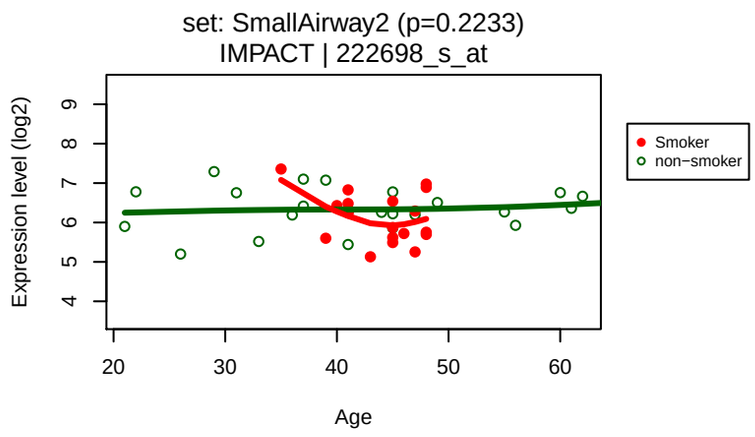
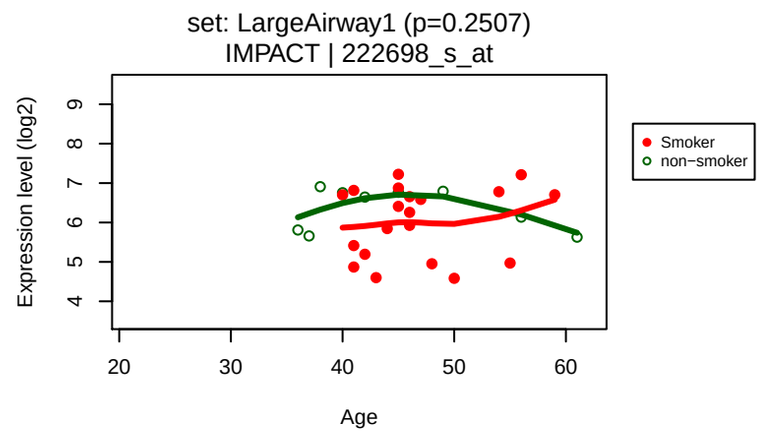

Gene expression of smokers and nonsmokers
gene: IMPACT | 222698_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3413
p=0.2233
p=0.4684
p=0.2507
p=0.0405
p=0.8931
p=0.3274
p=0.0824
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

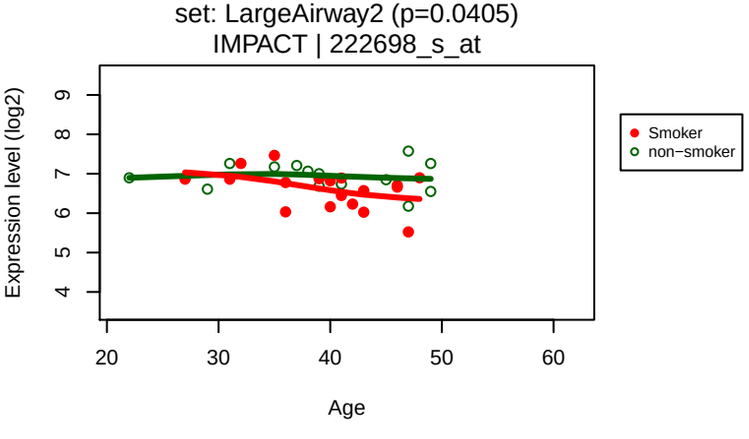

Gene expression ratio (smokers/nonsmokers)
gene: IMPACT | 222698_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3413
p=0.2233
p=0.4684
p=0.2507
p=0.0405
p=0.8931
p=0.3274
p=0.0824
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

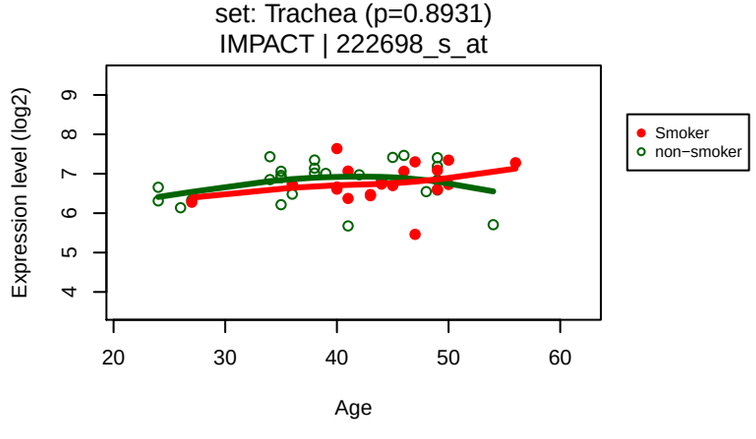

Gene expression ratio in Males
gene: IMPACT | 222698_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6627
p=0.0854
p=0.2783
p=0.2959
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IMPACT | 222698_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0224
p=0.4354
p=0.4204
p=0.2165
SmallAirway2
LargeAirway2
Trachea
Macrophage2

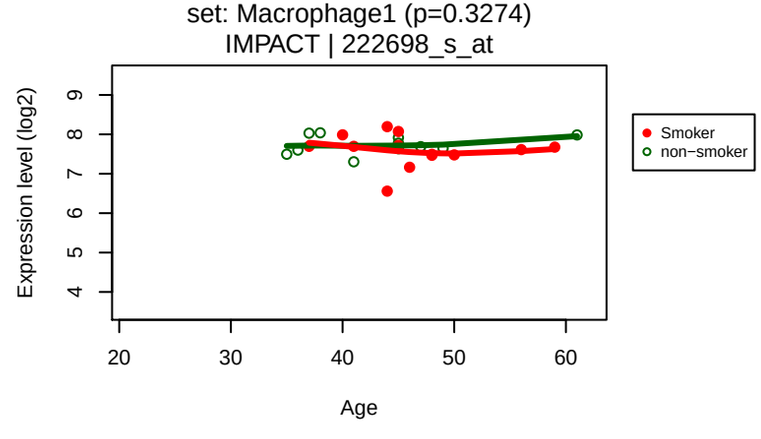

set: SmallAirway1 (p=0.3413)
IMPACT | 222698_s_at
Expression level (log2)
Age
Smoker
non−smoker

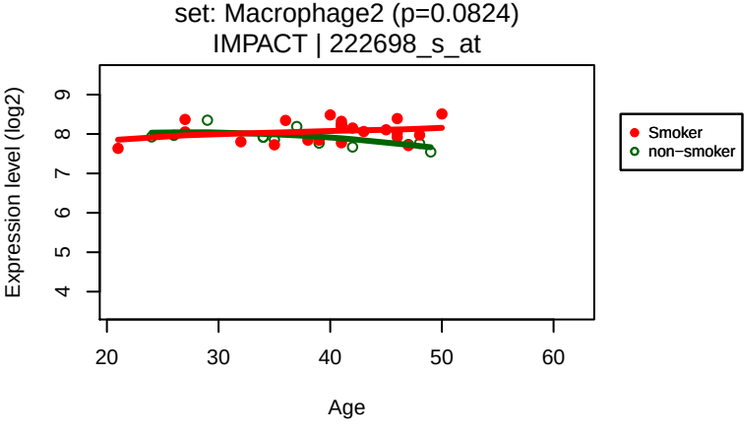

set: SmallAirway2 (p=0.2233)
IMPACT | 222698_s_at
Expression level (log2)
Age
Smoker
non−smoker

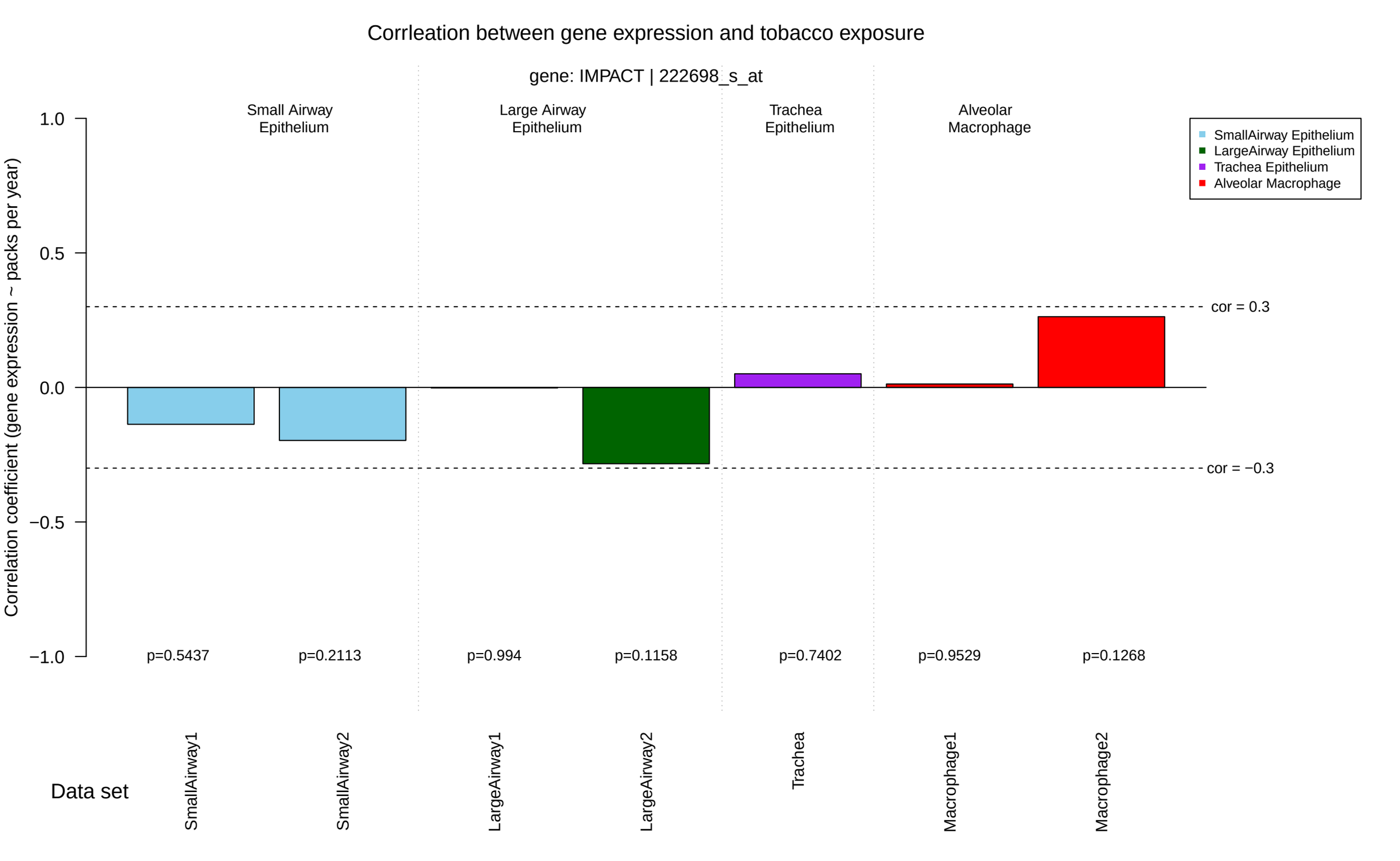

set: LargeAirway1 (p=0.2507)
IMPACT | 222698_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0405)
IMPACT | 222698_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8931)
IMPACT | 222698_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3274)
IMPACT | 222698_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0824)
IMPACT | 222698_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IMPACT | 222698_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.5437
p=0.2113
p=0.994
p=0.1158
p=0.7402
p=0.9529
p=0.1268
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2